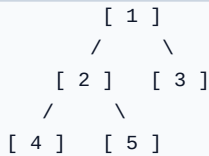


Binary Tree Traversals (Inorder, Preorder, Postorder, Level Order) -- Data Structures

Definition: Tree Traversal is the systematic process of visiting (reading, processing, or printing) every node in a tree data structure exactly once.

1. Detailed Technical Explanation



1. Depth-First Traversals (DFS):

1. Preorder Traversal (Root -> Left -> Right):

* Vis

2. Breadth-First Traversal (BFS / Level Order):

* Visits nodes level-by-level from top to bottom, and left to right at each level using a FIFO Queue.

* *E

2. Complete Executable C Program for Traversals

```
#include <stdio.h>
#include <stdlib.h>

struct Node {
    int data;
    struct Node* left;
    struct Node* right;
};

struct Node* createNode(int val) {
    struct Node* n = (struct Node*)malloc(sizeof(struct Node));
    n->data = val;
    n->left = n->right = NULL;
    return n;
}

// 1. Inorder Traversal: Left -> Root -> Right
void inorder(struct Node* root) {
    if (root != NULL) {
        inorder(root->left);
        printf("%d ", root->data);
        inorder(root->right);
    }
}
```

```
    }  
}  
  
// 2. Preorder Traversal: Root -> Left -> Right  
void preorder(struct Node* root) {  
    if (root != NULL) {  
        printf("%d ", root->data);  
        preorder(root->left);  
        preorder(root->right);  
    }  
}  
  
// 3. Postorder Traversal: Left -> Right -> Root  
void postorder(struct Node* root) {  
    if (root != NULL) {  
        postorder(root->left);  
        postorder(root->right);  
        printf("%d ", root->data);  
    }  
}  
  
int main() {  
    // Construct sample binary tree  
    struct Node* root = createNode(1);  
    root->left = createNode(2);  
    root->right = createNode(3);  
    root->left->left = createNode(4);  
    root->left->right = createNode(5);  
  
    printf("Preorder : "); preorder(root); printf("\n");  
    printf("Inorder  : "); inorder(root);  printf("\n");  
    printf("Postorder : "); postorder(root); printf("\n");  
    return 0;  
}
```

3. Reconstructing Unique Binary Tree from Traversals

- * A unique binary tree can be constructed if and only if INORDER is given along with either PREORDER or POSTORDER.

* Inc

4. Must-Write Points for Exams

- * Inorder traversal of BST gives non-decreasing sorted order.

* Tin

5. Quick Recall Flow

Preorder (V-L-R) Inorder (L-V-R) Postorder (L-R-V) Level Order (Queue BFS)
--